sysMonitor

Service Description

Service ID: *"sysMonitor"*

Abstract
This document describes the native `sysMonitor` service, version 1.0.0, produced by the PlantDescriptionEngine support system of Arrowhead (documentation edition 5.3.0).

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1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `sysMonitor` service, which provides native operational telemetry of the Plant DescriptionEngine system: periodic `echo`, `power` and `status` observations published to the local cloud's AMQP broker. The service is produced by PlantDescriptionEngine [1]; its concrete interface is the AMQP IDD [2].

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations.

2.1 Significant Prior Art

Arrowhead 4.x and upstream 5.2.2 had an HTTP `monitor/echo` endpoint. In this workspace the telemetry follows SysMonitor-AMQP-IDD-5.2.2; the HTTP liveness endpoint is not registered.

2.2 How This Service Is Meant to Be Used

The SysMonitor support system subscribes to the observations (selected with `sysMonitorManagement`) and keeps the history; LocalCloudManager shows it.

2.3 Important Delimitations and Dependencies

The observations describe the process, not the correctness of its answers. The service is offered only with `sysmonitor.enabled=true`. Delivery is publisher-confirmed but not end-to-end acknowledged by SysMonitor.

3 Service Interface

The `sysMonitor` service is a publishing service: the producer publishes one observation per operation (observation type) every publication interval (default 30 s); consumers receive them through the broker. Each operation has no input and the output Observation.

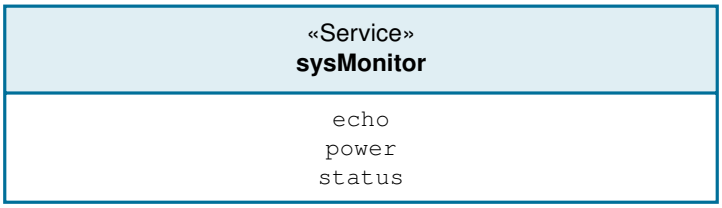


Figure 1: SysML-style block of the `sysMonitor` service and its observation types.

3.1 operation echo () : Observation

Liveness: `status` (`alive` or `degraded`), `uptimeSeconds` and a `detail` text; quality `good` or `uncertain`.

3.2 operation power () : Observation

Power supply: `supplyType=line`, `state=available`; unit `1`.

3.3 operation status () : Observation

Process state: `state=running`, `health` (`healthy`/`degraded`) and `serviceCount`, the number of registered services.

4 Information Model

4.1 struct Observation

One observation.

Field	Type	Description
schemaVersion	Version	Version of the observation schema, `1.0`.
observationId	Identifier	Unique identifier of the observation.
providerSystem	Name	The publishing system.
serviceInstance	Name	The service instance (default the system name).
observationType	Name	`echo`, `power`, `status`.
sourceTime	DateTime	Time of the observation.
sequence	Number	Sequence number per publisher.
value	Metadata	The observed value (per operation).
unit	Name	Unit of the value, if any.
quality	Name	`good` or `uncertain`.
metadata	Metadata	`publisherMode=native`.

4.2 Primitives

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “PlantDescriptionEngine – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-PlantDescriptionEngine-java-spring/documentation/v5.3.0/PlantDescriptionEngine-v5.3.0-SysD.pdf.

[2] ——, “sysMonitor AMQP/TLS/JSON – Interface Design Description (PlantDescriptionEngine),” Eclipse Arrowhead, IDD 5.3.0, 2026, plantDescriptionEngine-v5.3.0-sysMonitor-AMQP-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-23	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	